

CASE STUDY

PhonePe Payment Insights Dashboard

Turning 300K+ digital transactions into business-ready insights using Power BI, DAX, and data storytelling

Role	Data Analyst (Independent Project)
Tools Used	Power BI Desktop, Power Query, DAX, Excel/CSV
Dataset	PhonePe transaction data — 300K transactions, 100.76K unique users
Project Type	Self-initiated learning project
GitHub Repo	github.com/rahulplots/PhonePe-Analytics-Dashboard

Overview

This project analyzes transaction-level data from a digital payments platform to understand how users transact, which services drive the most value, and how reliable the payment ecosystem is. The goal was to go beyond just building charts — to ask real business questions and answer them with data, the way an analyst would for an actual fintech product team.

Every KPI, DAX measure, and visual in this dashboard was designed and built independently, with a focus on dashboard storytelling rather than just displaying numbers.

Business Problem

A digital payments platform processes thousands of transactions every day across multiple services — bill payments, recharges, money transfers, insurance, and loans. Without a structured way to track this data, it becomes difficult for stakeholders to answer basic but critical questions:

- How is transaction activity trending month over month?
- Which services actually generate the most value for the business?
- Are payments failing often enough to be a reliability concern?
- Which customer segments are driving the bulk of platform usage?
- Are there usage patterns (like weekday vs weekend) the business should plan around?

The objective of this project was to build a single interactive dashboard that could answer all of these questions at a glance — turning raw transaction logs into decisions a product or business team could actually act on.

Approach

1. Data Preparation

Raw transaction data was loaded and cleaned using Power Query, handling formatting and structuring it for relational modeling inside Power BI before building any visuals.

2. Metric & KPI Design

Designed DAX measures from scratch to calculate total transaction value, month-over-month growth, payment success rate, and segment-wise contribution — rather than relying on raw columns.

3. Dashboard Architecture

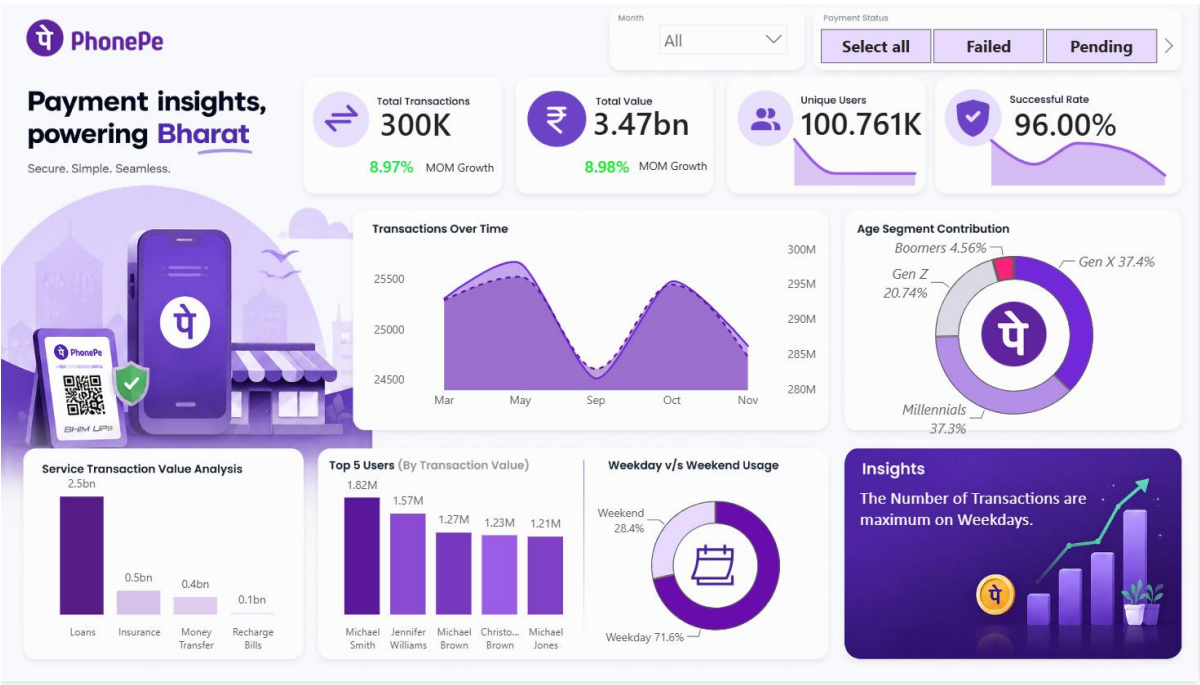
The dashboard was structured around five focus areas so any viewer could navigate from a high-level summary down to specific behavior patterns:

- Executive Summary — headline KPIs (transactions, value, users, success rate)
- Transaction Analysis — monthly trends and growth tracking
- Customer Insights — age-wise segmentation
- Service Performance — value comparison across service categories
- Usage Behaviour — weekday vs weekend activity, top users by value

4. Storytelling Layer

Custom tooltips were built for deeper context on hover, and a dedicated insights panel was added so the dashboard doesn't just show data — it explains what the data means.

Dashboard Preview



Full interactive Power BI dashboard — Executive Summary view

Key Performance Indicators

300K Total Transactions 8.97% MoM	₹3.47Bn Transaction Value 8.98% MoM	100.76K Unique Users	96.0% Success Rate
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Key Insights

- **Loan services generate the highest transaction value**
Loan-related transactions contribute approximately ₹2.5 billion — making loans the single most valuable service category on the platform, far ahead of insurance, money transfer, and recharge/bill payments.
- **Strong payment reliability**
The platform maintains a 96% payment success rate, indicating a dependable transaction pipeline with relatively low failure overhead.
- **Weekdays drive most platform activity**

71.6% of all transactions happen on weekdays versus 28.4% on weekends — usage is tied closely to routine, work-week financial activity rather than leisure-time spending.

- **Millennials and Gen X are the core user base**

These two segments together account for nearly 75% of platform usage, making them the most important segments to design future features and offers around.

- **A small group of high-value users has outsized impact**

The top users on the platform individually generate transaction values exceeding ₹1 million, highlighting how much retaining a small premium segment matters to overall revenue.

- **Healthy, consistent growth**

Both transaction volume and transaction value show a positive month-over-month trend, suggesting the platform is gaining sustained usage rather than one-off spikes.

Business Recommendations

- Continue investing in loan-related services, since they are the platform's strongest value driver.
- Introduce weekend-specific offers to lift the comparatively low weekend transaction share.
- Focus marketing and product efforts on Millennial and Gen X segments, who already drive most usage.
- Build loyalty or retention programs targeted at high-value/premium users.
- Continue monitoring the payment pipeline to maintain the 96% success rate as volume scales.

What I Learned

This project pushed me to think like an analyst solving a business problem, not just someone building charts. A few specific takeaways:

- Designing custom tooltips for the first time — and realizing how much they improve how a dashboard communicates context.
- Writing DAX measures for business calculations like MoM growth and success rate, not just basic aggregations.
- Planning dashboard layout around questions a stakeholder would actually ask, instead of around the data I happened to have.
- The biggest lesson: a good dashboard isn't about how many visuals it has — it's about whether someone can look at it and immediately understand what's happening in the business.

Tech Stack

- Power BI Desktop — dashboard development and interactive reporting
- Power Query — data cleaning and transformation
- DAX — custom measures, KPIs, and business calculations
- Excel / CSV — source data handling

Links

GitHub Repository: github.com/rahulplots/PhonePe-Analytics-Dashboard

Portfolio: rahulplots.github.io

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